

화학수학

열한 번째 수업

미분방정식

Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
 - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$

Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions

(일반해와 특수해)

- **Solution:** any function that reduces the DE to an identity when substituted into the DE

For a DE of order n ,

- **General solution:** contains n arbitrary constants
- **Particular solution:** contains no arbitrary constants
 - Particular solutions can be generated from the general solution



Additional Conditions

- For a DE of order n , if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first $(n - 1)$ derivatives at a single point
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Solving a Differential Equation

1. Obtain the **general solution**: hardest part!
 2. If initial/boundary conditions are given, substitute the conditions to **determine the coefficients**
- No universal method to solve all DEs
 - For specific forms of DEs, we have general solving methods

First-Order ODEs

$$F\left(y, \frac{dy}{dx}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y) \frac{dy}{dx} = f(x)$$

This DE can be manipulated into

$$g(y) dy = f(x) dx$$

$$\int g(y) dy = \int f(x) dx + C$$

This equation provides the **general solution**



Exact Differential Equations

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **exact**,
there is a function $f(x, y)$ such that

$$df = M(x, y) dx + N(x, y) dy = 0$$

which implies that $f(x, y) = k$ (constant)

How to Find $f(x, y)$

$$df = M(x, y) dx + N(x, y) dy = 0$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_C df$$

where C is a curve connecting (x_0, y_0) and (x_1, y_1) . Take a rectangular path from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) dx + \int_{y_0}^{y_1} N(x_1, y) dy$$

This gives the relationship between x and y .

Integrating Factors

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **inexact**, but we know the **integration factor** $g(x, y)$ that makes

$$g(x, y)M(x, y) dx + g(x, y)N(x, y) dy = 0$$

an exact differential equation, we can solve this DE.

Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{dy}{dx} + P(x)y = f(x)$$

- General solution:

$$y = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx + C e^{-\int P(x)dx}$$

Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
 - Homogeneous (§ 12.2)
 - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs



Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \dots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \dots, c_k$.



Reduction of Order

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution y_1 is known,
we can obtain the other solution y_2

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

오늘의 수업

- Homogeneous linear ODEs with constant coefficients
 - Solving strategy
 - Application: harmonic oscillator
 - Application: particle in a box
- Heterogeneous linear ODEs with constant coefficients
 - Solving strategy
 - Application: harmonic oscillator



Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Homogeneous Case

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시험해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Roots of Characteristic Equation

$$a\lambda^2 + b\lambda + c = 0$$

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

1. Two roots are real and distinct ($b^2 - 4ac > 0$)
2. Two roots are real and equal ($b^2 - 4ac = 0$)
3. Two roots are complex numbers ($b^2 - 4ac < 0$)

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

Example 12.2

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0.$$

The characteristic equation is

$$\lambda^2 + \lambda - 2 = 0$$

whose solutions are

$$\lambda_1 = 1, \lambda_2 = -2$$

Hence, the general solution is

$$y = c_1 e^x + c_2 e^{-2x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0.$$

The characteristic equation is

$$\lambda^2 - 10\lambda + 25 = 0$$

whose solution is

$$\lambda = 5$$

Hence, the general solution is

$$y = c_1 e^{5x} + c_2 x e^{5x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x}$$

(3) Conjugate Complex Roots

$$y = c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x}$$

Since $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$,

$$\begin{aligned} y &= c_1 e^{\alpha x} (\cos \beta x + i \sin \beta x) + c_2 e^{\alpha x} (\cos \beta x - i \sin \beta x) \\ &= e^{\alpha x} \left[\underbrace{(c_1 + c_2)}_{c'_1} \cos \beta x + i \underbrace{(c_1 - c_2)}_{c'_2} \sin \beta x \right] \end{aligned}$$

$$y = e^{\alpha x} (c'_1 \cos \beta x + c'_2 \sin \beta x)$$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 7y = 0.$$

The characteristic equation is

$$\lambda^2 + 4\lambda + 7 = 0$$

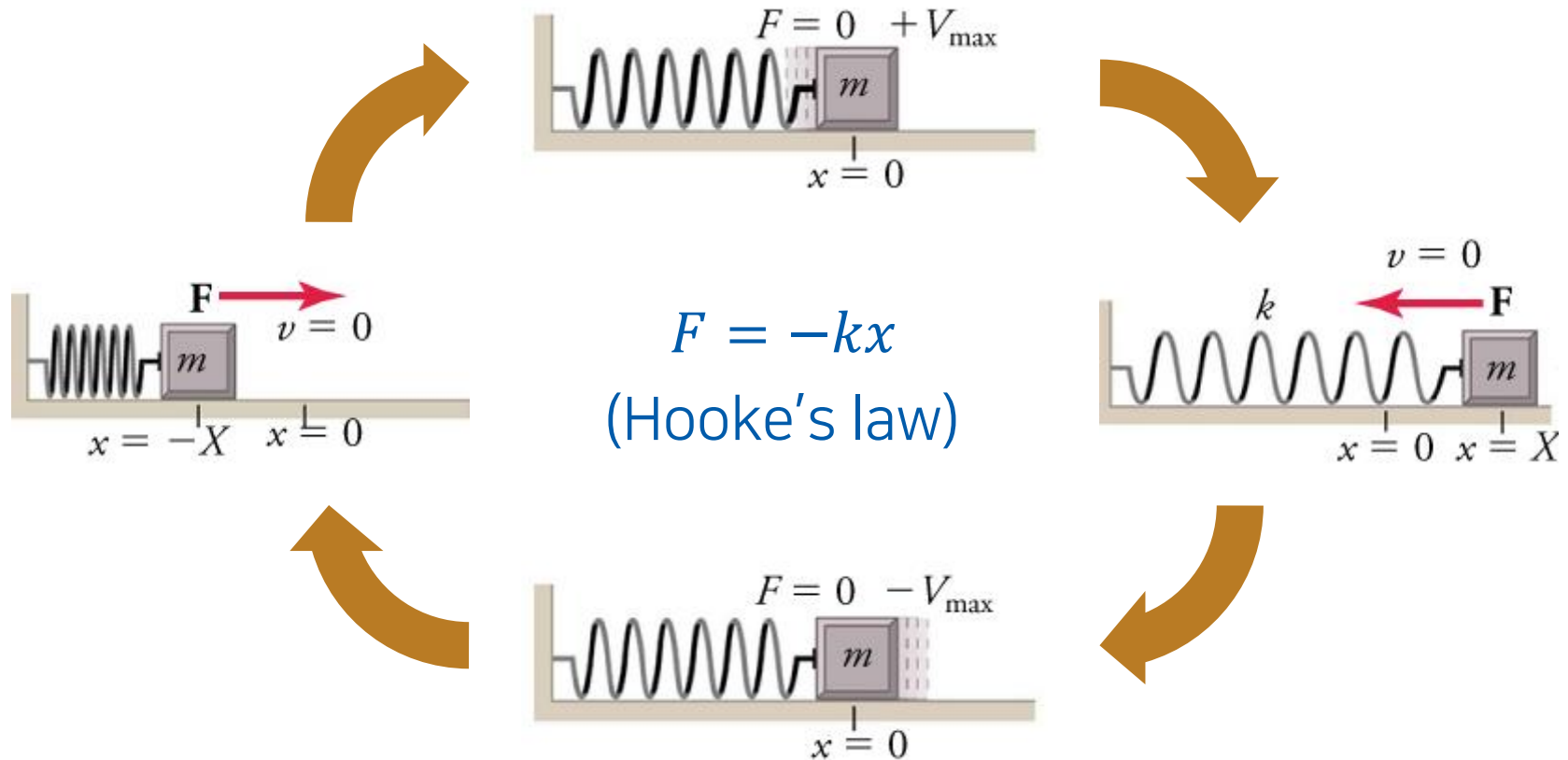
whose solutions are

$$\lambda_1 = -2 + i\sqrt{3}, \lambda_2 = -2 - i\sqrt{3}$$

Hence, the general solution is

$$y = e^{-2x} [c_1 \cos(\sqrt{3}x) + c_2 \sin(\sqrt{3}x)]$$

Application: Harmonic Oscillator (조화진동자)



(Image: <https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion>)

Equation of Motion (운동방정식)

- Newton's second law

$$F = ma = m \frac{dv}{dt} = m \frac{d^2x}{dt^2}$$

- Equation of motion

$$F = m \frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Solving Equation of Motion

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

The characteristic equation is

$$\lambda^2 + \frac{k}{m} = 0$$

whose solutions are

$$\lambda_1 = i\sqrt{k/m}, \quad \lambda_2 = -i\sqrt{k/m}$$

Solving Equation of Motion

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\lambda_1 = i\sqrt{k/m}, \quad \lambda_2 = -i\sqrt{k/m}$$

Define $\omega = \sqrt{k/m}$; the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Assume that we know the conditions at $t = 0$:

$$x(0) = 0, \quad v(0) = \left. \frac{dx}{dt} \right|_{t=0} = v_0$$

From the first condition,

$$x(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 0$$

$$x(t) = c_2 \sin \omega t$$

Initial Conditions

$$x(t) = c_2 \sin \omega t$$

$$v(0) = \left. \frac{dx}{dt} \right|_{t=0} = v_0$$

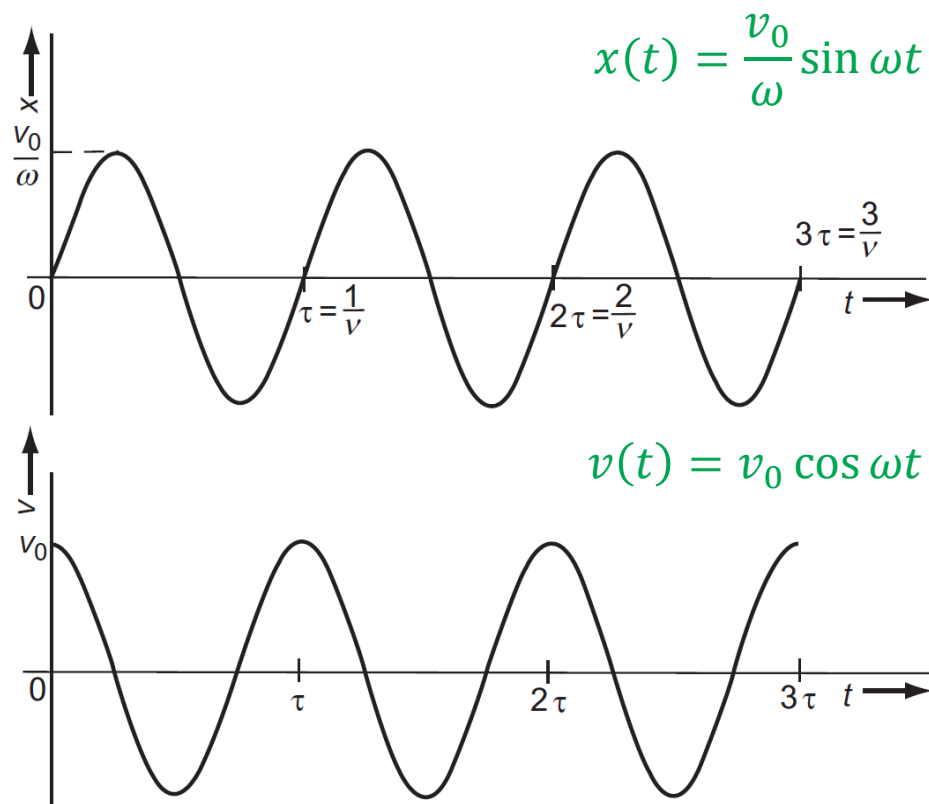
From the second condition,

$$v(0) = c_2 \omega \cos 0 = c_2 \omega = v_0$$

Hence, the solution is

$$x(t) = \frac{v_0}{\omega} \sin \omega t$$

Uniform Harmonic Motion

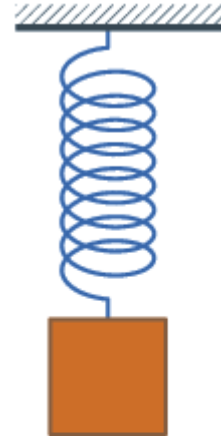


(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.1)

Damped Harmonic Oscillator

(감쇠 조화진동자)

- Damping (감쇠): reduction of oscillation
- Application of **friction force** (마찰력)
 - proportional to the velocity
 - in the opposite direction



$$\vec{F}_f = -\zeta \vec{v} = -\zeta \frac{d\vec{r}}{dt}$$

friction constant
(마찰 상수)

1-Dimensional Equation of Motion

$$F = m \frac{d^2 x}{dt^2} = -kx - \zeta \frac{dx}{dt}$$

- Standard form

$$\frac{d^2 x}{dt^2} + \frac{\zeta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

- Characteristic equation

$$\lambda^2 + \frac{\zeta}{m} \lambda + \frac{k}{m} = 0$$

Roots of Characteristic Equation

$$\lambda^2 + \frac{\zeta}{m}\lambda + \frac{k}{m} = 0$$

$$D = \left(\frac{\zeta}{m}\right)^2 - \frac{4k}{m}$$

1. Distinct real roots ($D > 0$)
2. Repeated real roots ($D = 0$)
3. Conjugate complex roots ($D < 0$)

(1) $D > 0$: Overdamping (과감쇠)

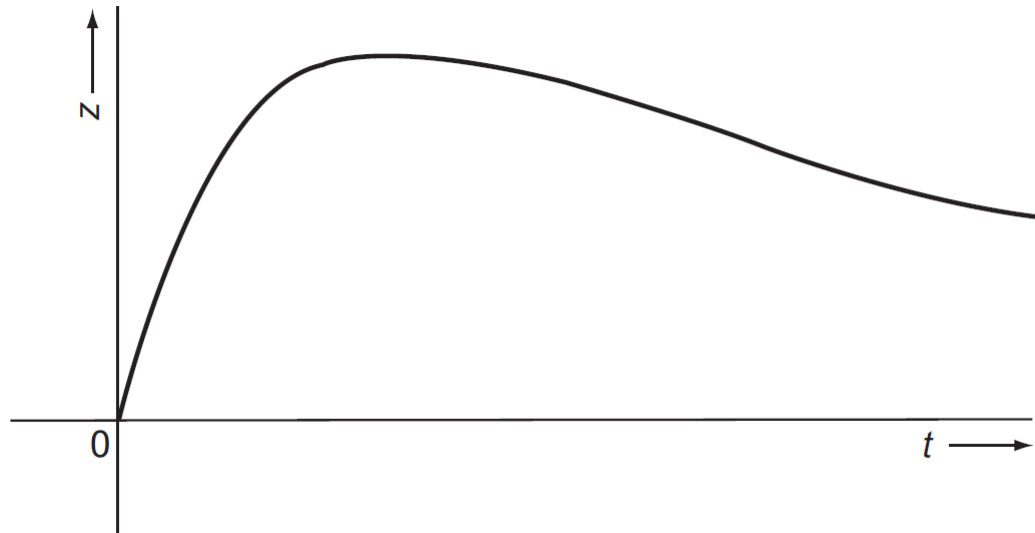
$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4km}}{2m}, \quad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4km}}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

- Note that $\lambda_1, \lambda_2 < 0$ (Example 12.6) → decay!
- If the initial condition is $x(0) = 0$, $c_1 = -c_2$

(1) $D > 0$: Overdamping (과감쇠)

$$x(t) = c(e^{\lambda_1 t} - e^{\lambda_2 t})$$



(2) $D = 0$: Critical Damping (임계감쇠)

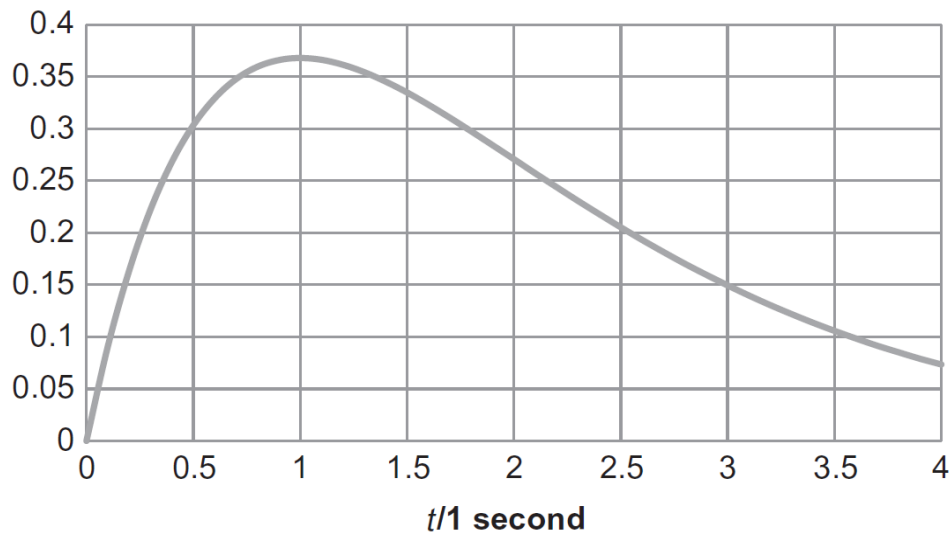
$$\lambda_1 = -\frac{\zeta}{2m}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t}$$

- Note that $\lambda_1 < 0 \rightarrow$ decay!
- If the initial condition is $x(0) = 0$, $c_1 = 0$

(2) $D = 0$: Critical Damping (임계감쇠)

$$x(t) = c_2 t e^{\lambda_1 t}$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.4)

(3) $D < 0$: Underdamping (미급감쇠)

$$\lambda_1 = \frac{-\zeta + \sqrt{\zeta^2 - 4km}}{2m}, \quad \lambda_2 = \frac{-\zeta - \sqrt{\zeta^2 - 4km}}{2m}$$

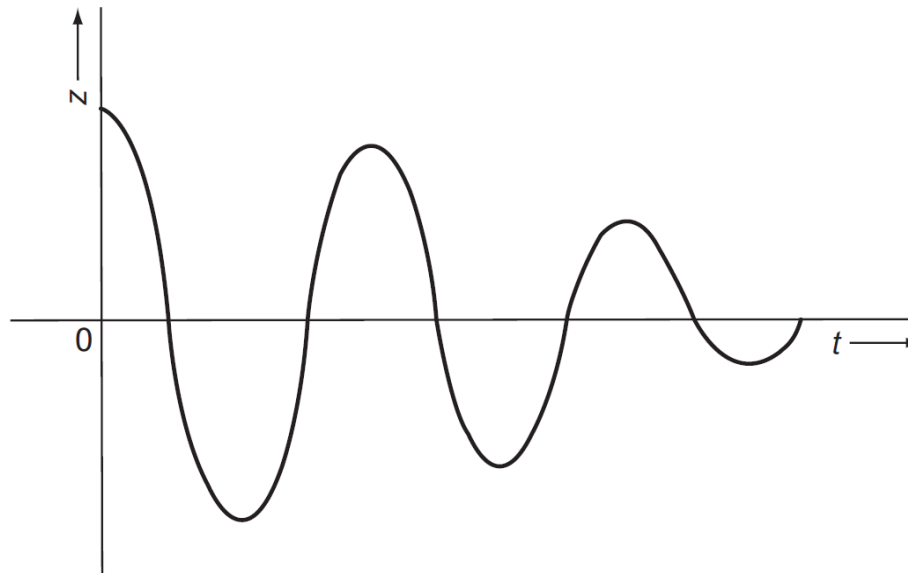
Let $\omega = \sqrt{(k/m) - (\zeta/2m)^2}$;

$$\lambda_1 = -\frac{\zeta}{2m} + i\omega, \quad \lambda_2 = -\frac{\zeta}{2m} - i\omega$$

$$x(t) = e^{-\zeta t/2m} (c_1 \cos \omega t + c_2 \sin \omega t)$$

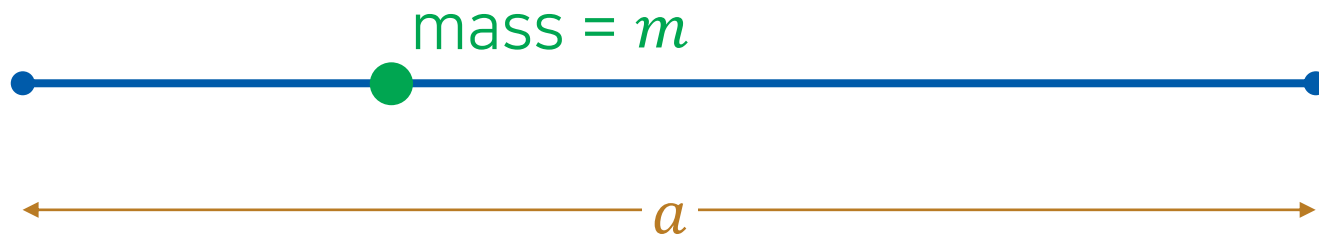
(3) $D < 0$: Underdamping (미급감쇠)

$$x(t) = ce^{-\zeta t/2m} \cos \omega t$$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.3)

Application: Particle in a Box



$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

Boundary conditions: $\psi(0) = \psi(a) = 0$

Solving the Differential Equation

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

- Characteristic equation: $\lambda^2 + \frac{2mE}{\hbar^2} = 0$

Since $E > 0$, the characteristic equation has two conjugate complex roots:

$$\lambda_1 = \frac{i\sqrt{2mE}}{\hbar}, \quad \lambda_2 = -\frac{i\sqrt{2mE}}{\hbar}$$

General Solution

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

$$\lambda_1 = \frac{i\sqrt{2mE}}{\hbar}, \quad \lambda_2 = -\frac{i\sqrt{2mE}}{\hbar}$$

Hence, the general solution is

$$\psi(x) = c_1 \cos\left(\frac{\sqrt{2mE}}{\hbar}x\right) + c_2 \sin\left(\frac{\sqrt{2mE}}{\hbar}x\right)$$

Boundary Conditions

$$\psi(x) = c_1 \cos\left(\frac{\sqrt{2mE}}{\hbar} x\right) + c_2 \sin\left(\frac{\sqrt{2mE}}{\hbar} x\right)$$

- **Boundary conditions:** $\psi(0) = \psi(a) = 0$

$$\psi(0) = c_1 \cos 0 + c_2 \sin 0 = c_1 = 0$$

$$\psi(a) = c_2 \sin\left(\frac{\sqrt{2mE}}{\hbar} a\right) = 0$$

To avoid $c_2 = 0$, we obtain $\frac{\sqrt{2mE}}{\hbar} a = n\pi$ (n : integer)

Final Solution

$$\frac{\sqrt{2mE}}{\hbar} a = n\pi$$

$$E = \frac{\hbar^2 \pi^2}{2ma^2} n^2$$

The final solution is

$$\psi(x) = c_2 \sin\left(\frac{n\pi x}{a}\right)$$

(One can use normalization to determine c_2 .)

Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = \underline{g(x)}$$

inhomogeneous term
(비동차항)

- Complementary equation (보상방정식)

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

Solving Inhomogeneous ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation →
complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_c + y_p$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 16x.$$

A particular solution to this equation is $y = 4x + 5$.

The complementary equation is

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 0$$

and the complementary function is

$$y_c(x) = c_1e^x + c_2e^{4x}$$

Hence, the general solution is

$$y = c_1e^x + c_2e^{4x} + 4x + 5$$

Method of Undetermined Coefficients

(미정계수법)

How can we obtain a particular solution y_p ?

Guess the form of y_p with undetermined coefficients!

Trial Solutions

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$A e^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t} (A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 2x^2 - 3x + 6.$$

The complementary equation is

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$$

and the complementary function is

$$y_c(x) = c_1e^{-2x} + c_2xe^{-2x}$$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 2x^2 - 3x + 6.$$

$$y_c(x) = c_1e^{-2x} + c_2xe^{-2x}$$

Now, consider a trial solution $y_p = A_0 + A_1x + A_2x^2$, whose derivatives are

$$\frac{dy_p}{dx} = A_1 + 2A_2x, \quad \frac{d^2y_p}{dx^2} = 2A_2$$

Substitution leads to

$$4A_2x^2 + (8A_2 + 4A_1)x + (4A_0 + 4A_1 + 2A_2) = 2x^2 - 3x + 6$$

Comparison gives $A_2 = 1/2, A_1 = -7/4, A_0 = 3$.

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 2x^2 - 3x + 6.$$

$$y_c(x) = c_1e^{-2x} + c_2xe^{-2x}$$

$$y_p(x) = \frac{1}{2}x^2 - \frac{7}{4}x + 3$$

The general solution is

$$y = c_1e^{2x} + c_2xe^{2x} + \frac{1}{2}x^2 - \frac{7}{4}x + 3$$

Glitch in the Method

A trial solution y_p can be
a solution of the complementary equation.

In this case, try $x^n y_p$ instead!

(n : smallest integer that resolves the problem)

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8e^x.$$

The complementary function is

$$y_c(x) = c_1e^x + c_2e^{4x}$$

According to Table 12.1, the trial solution for $8e^x$ is Ae^x .

However, substitution gives

$$Ae^x - 5Ae^x + 4Ae^x = 0 = 8e^x$$

The trial solution is a wrong guess.

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8e^x.$$

$$y_c(x) = c_1e^x + c_2e^{4x}$$

Instead of Ae^x , we can try $y_p = Axe^x$.

$$\frac{dy_p}{dx} = Ae^x + Axe^x, \quad \frac{d^2y_p}{dx^2} = 2Ae^x + Axe^x$$

Hence,

$$\frac{d^2y_p}{dx^2} - 5\frac{dy_p}{dx} + 4y_p = -3Ae^x$$

 and $A = -8/3$.

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 4y = 8e^x.$$

$$y_c(x) = c_1e^x + c_2e^{4x}$$

$$y_p(x) = -\frac{8}{3}xe^x$$

Therefore, the general solution is

$$y(x) = c_1e^x + c_2e^{4x} - \frac{8}{3}xe^x$$